Md. Mutasim Billah Abu Noman Akanda

PhD Applicant (Fall 2027) • Document AI and Low-Resource OCR

+880-1773272649 Website Email LinkedIn GitHub Scholar Kaggle

Research Interests

Document AI and OCR for low-resource languages (layout analysis, handwriting recognition, layout-aware evaluation); efficient multimodal / VLM adaptation under deployment budgets; biomedical image analysis (MRI/CT DICOM pipelines); reproducible multi-GPU training and metric design. Current focus: Bengali handwritten OCR with parse-success-gated text-word-layout scoring, sweep-guided model selection, and 4-GPU DDP workflows.

Education

- **University of Liberal Arts Bangladesh (ULAB)** Dhaka, Bangladesh
  B.Sc. Computer Science and Engineering *Summer 2019 – Fall 2022*

  CGPA: 3.96/4.00 **Distinction:** Summa Cum Laude (top 1%; conferred May 2024)
  Honors: Vice Chancellor's Honors List; Dean's List Scholarship (3×)
  Relevant coursework: Artificial Intelligence, Digital Image Processing, Algorithms, Data Structures, Statistics and Probability, Software Engineering

Publications

- **Akanda, M.B.A.N.*, Ahmed, M.*, Rabby, A.S.A., & Rahman, F.**: *Optimum Deep Learning Method for Document Layout Analysis in Low Resource Languages.* ACM Southeast Conference (ACM SE'24), 199–204. doi:10.1145/3603287.3651184. *Equal contribution.
- **Akanda, M.B.A.N., Prodhan, M., Sarwar, S., Raatul, A.M., Paul, B.**: *Voice Controlled Home Automation with Cloud-Based Environment Monitoring System.* ICTCS 2022, Lecture Notes in Networks and Systems, vol. 623, Springer, Singapore. doi:10.1007/978-981-19-9638-2_21

Research and Professional Experience

- **Apurba Technologies Ltd.** Dhaka, Bangladesh
  Senior Machine Learning Engineer *Sep 2025 – present*
  - **Current research project (ongoing)**: Leading Bengali handwriting OCR fine-tuning with a reproducible workflow: strict JSONL validation, W&B sweep screening, and 4-GPU DDP full training for final model selection.
  - **Objective-aligned evaluation**: Using a joint text-word-layout optimization target ('eval_text_word_layout_score') with parse-success gating to prioritize both textual correctness and structural fidelity.
  - **Training-system rigor**: Implemented sweep-equivalent global batch behavior across GPU counts and DDP-safe OCR metrics controls to preserve comparability between screening runs and full training.
  - **Research relevance**: Converting production constraints (layout variance, OCR noise, memory limits) into measurable experiments on robustness, efficiency, and deployment reliability.
- **Apurba Technologies Ltd.** Dhaka, Bangladesh
  Machine Learning Engineer *Mar 2023 – May 2024*
  - **Low-resource document layout analysis**: Built and evaluated YOLOv8-based layout models for Bengali document structures (**85%+** internal accuracy), supporting OCR quality improvements in deployment.
  - **Inference optimization**: Implemented 8-bit quantization and measured **2**× faster inference with reduced memory usage, enabling broader operational deployment.
  - **Research dissemination**: Co-first author on ACM SE'24 Bengali DLA study (BADLAD); co-author on Springer ICTCS 2022 paper.
- **Pixalate Inc.** Remote (California, USA)
  Data Scientist *Mar 2025 – Aug 2025*
  - **Web-scale ML systems**: Designed AI-assisted ad-network detection workflows over large crawling pipelines (**2M+** sites), combining learned and deterministic methods.
  - **Methodological trade-off analysis**: Used hybrid detection strategies to balance precision, throughput, and infrastructure cost in production settings.
- **Green Pants Studio** Remote (Texas, USA)
  AI Engineer *Jun 2024 – May 2025*

- **Applied ML for valuation and mapping**: Developed ML-supported product valuation and data mapping workflows for large eCommerce streams.
- **Pipeline quality and reliability**: Improved scraper throughput and deployment reliability through modular services and containerized delivery.

Teaching Experience

- **University of Liberal Arts Bangladesh (ULAB)** — Dhaka, Bangladesh
  Teaching Assistant, Department of Computer Science and Engineering — *Feb 2021 – Jan 2023*
  - **Courses supported**: Introduction to Programming, Object-Oriented Programming, Artificial Intelligence, Software Engineering
  - **Instructional duties**: Led laboratory sessions and tutorials for cohorts of **30+** students; prepared lab materials and evaluated assignments/projects.
  - **Mentoring**: Guided students on programming and introductory AI/ML projects during office hours and practical sessions.

Leadership

- **ULAB Computer Programming Club** — Dhaka, Bangladesh
  President — *2022 – 2023*
  - **Contest organization**: Led **4** UCPC competitive programming contests (2022–2023), including Take Off Summer 2022 (**40+** participants).
  - **Problem setting**: Lead problem setter, Take Off Summer 2022 (**6**-problem set; ULAB official announcement).

Biomedical Imaging (Kaggle / RSNA)

- **RSNA-MICCAI Brain Tumor Radiogenomic Classification**: Built PyTorch EfficientNet3D pipelines on mpMRI DICOM (FLAIR, T1w, T1wCE, T2w) for MGMT promoter methylation classification; middle-slice volume stacking and DICOM preprocessing with VOI LUT. Kaggle
- **RSNA 2023 Abdominal Trauma Detection**: Explored CT trauma detection with PyTorch 2.5D EfficientNet CNN (5-fold CV), 3D R3D-18 on DICOM slices, KerasCV CNN training, and DICOM-to-3D tensor preprocessing. Kaggle

Selected Projects (Research-Relevant)

- **SenseScan (Bengali handwritten OCR)**: Pipeline with EAST/LANMS detection, CRNN recognition, and reading-order assembly; designed for experimentation in low-resource OCR settings. GitHub
- **Voice Healthcare Agent**: Multimodal conversational pipeline (FastAPI + Next.js + local voice stack) useful for studying latency/reliability trade-offs in speech-enabled LLM systems. GitHub
- **GradConnectAI**: Evidence-based matching system for graduate supervision discovery; relevant to ranking/retrieval experimentation in academic recommendation settings. GitHub

Reproducibility and Tooling

- **Experiment support**: Use MLflow/Weights & Biases style tracking and test-driven workflows (pytest/Vitest) to improve repeatability and regression control in ML pipelines.
- **Deployment-aware reproducibility**: Maintain API-first project structures (FastAPI, Docker, CI) to align research prototypes with executable, verifiable systems.
- **Data and evaluation mindset**: Prioritize clear evaluation criteria, metric definitions, and error-analysis loops when moving from prototype to production conditions.

Technical Skills

- **Languages**: Python, TypeScript, SQL
- **ML/AI**: PyTorch, TensorFlow, scikit-learn, Hugging Face, OpenCV, quantization workflows
- **Medical imaging**: pydicom, DICOM preprocessing, EfficientNet3D, 3D/video CNNs (R3D-18), KerasCV
- **LLM/NLP**: RAG pipelines, vector search (pgvector/ChromaDB), vLLM, Ollama
- **Systems**: FastAPI, Docker, PostgreSQL, Redis, SQLite, async crawling and data pipelines
- **Experimentation**: MLflow, Weights & Biases, pytest, Vitest

Honors and Awards

- **Summa Cum Laude**: Top 1% of graduating class (May 2024)
- **Vice Chancellor's Honors List**: Scholarship recipient
- **Dean's List Scholarship**: Awarded 3 times
- **Programming contests**: Problem setter and award recipient in ULAB Take Off contests
- **ULAB Computer Programming Club**: President (2022–2023); organized **4** contests; Take Off Summer 2022 (**40+** participants)